

Deep Learning

Lecture 1: Introduction

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Lecture overview a framework for your deep learning armamentarium



1 Introduction

- definitions
- examples

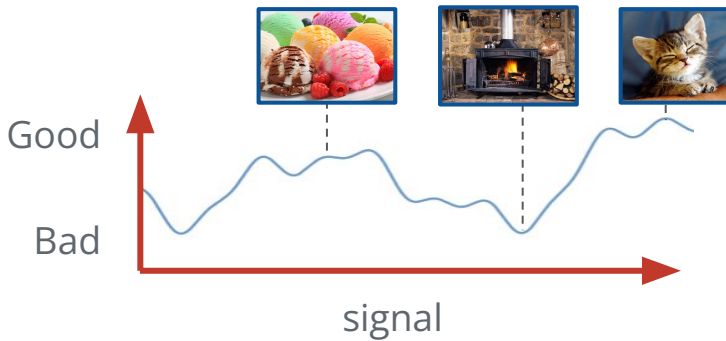
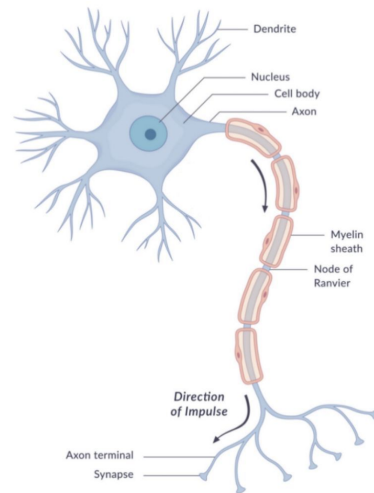
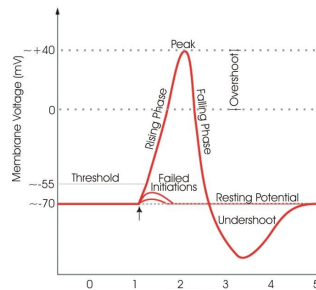
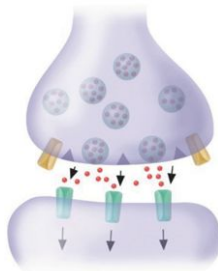
2 Learning in Nature

- what is learning?
- how does the brain work?
- synaptic plasticity
- Hebbian theory

3 A Brief History

4 Key Concepts

- where it fits in to ML/RL
- three spaces



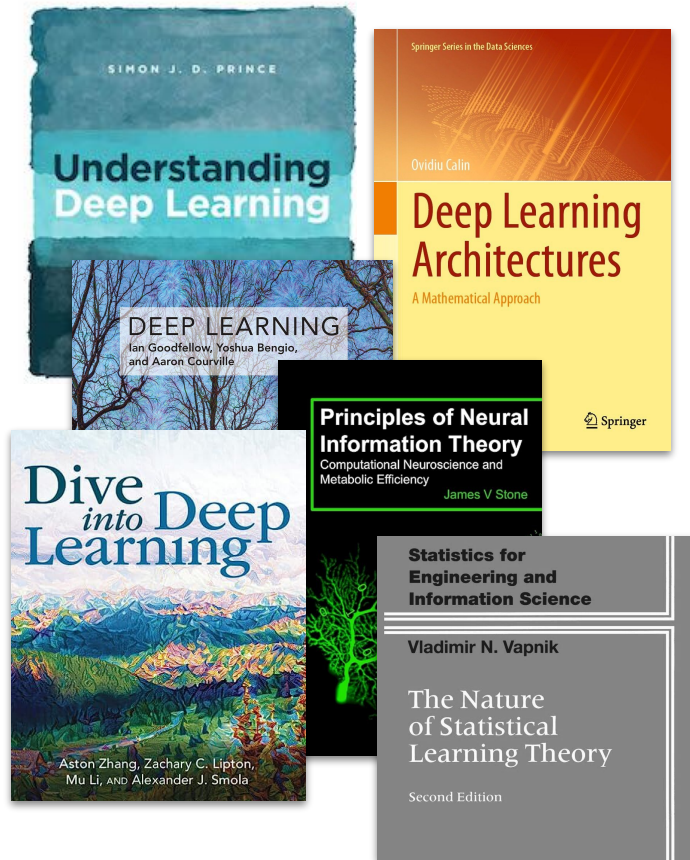
The aims of the course are:

1. To be able to solve complex ill-defined problems that require deep layers of learning
2. To understand natural learning, and how it connects to the deep learning theory
3. To understand and be able to implement the foundations of state-of-the-art papers in the engineering and research communities

 PyTorch



colab

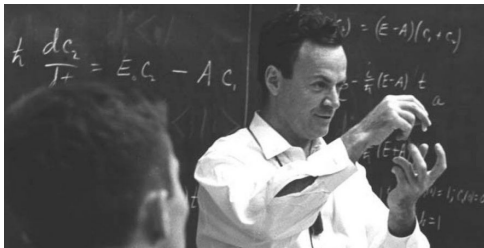




Definition: Learning

"We define learning as the transformative process of taking in information that—when internalized and mixed with what we have experienced—changes what we know and builds on what we do. It's based on input, process, and reflection. It is what changes us."

Tony Bingham and Marcia Conner





Definition: Deep Learning

“Deep learning allows computational models that are composed of multiple processing layers to learn representations of data with multiple levels of abstraction. These methods have dramatically improved the state-of-the-art in speech recognition, visual object recognition, object detection and many other domains such as drug discovery and genomics. Deep learning discovers intricate structure in large data sets by using the backpropagation algorithm.”

Yann LeCun, Yoshua Bengio & Geoffrey Hinton
Deep Learning, 2015, Nature - [Read Online](#)

On the Number of Linear Regions of Deep Neural Networks (Montúfar et al.)
The Power of Depth for Feedforward Neural Networks (Eldan et al.)



Before: Images and Vision

Non-photorealistic interpolation:

- Photos - [video](#)
- Paintings - [video](#)
- DeepFakes - [video](#)

End-to-end self-driving:

- Wayve - [video](#)

Now: Multimodality tokenization

Examples from text:

- GPT-6 and **multimodality**
- **Input:** The internet, reasoning
- **Output:** Multimodal tasks

Examples with audio and control:

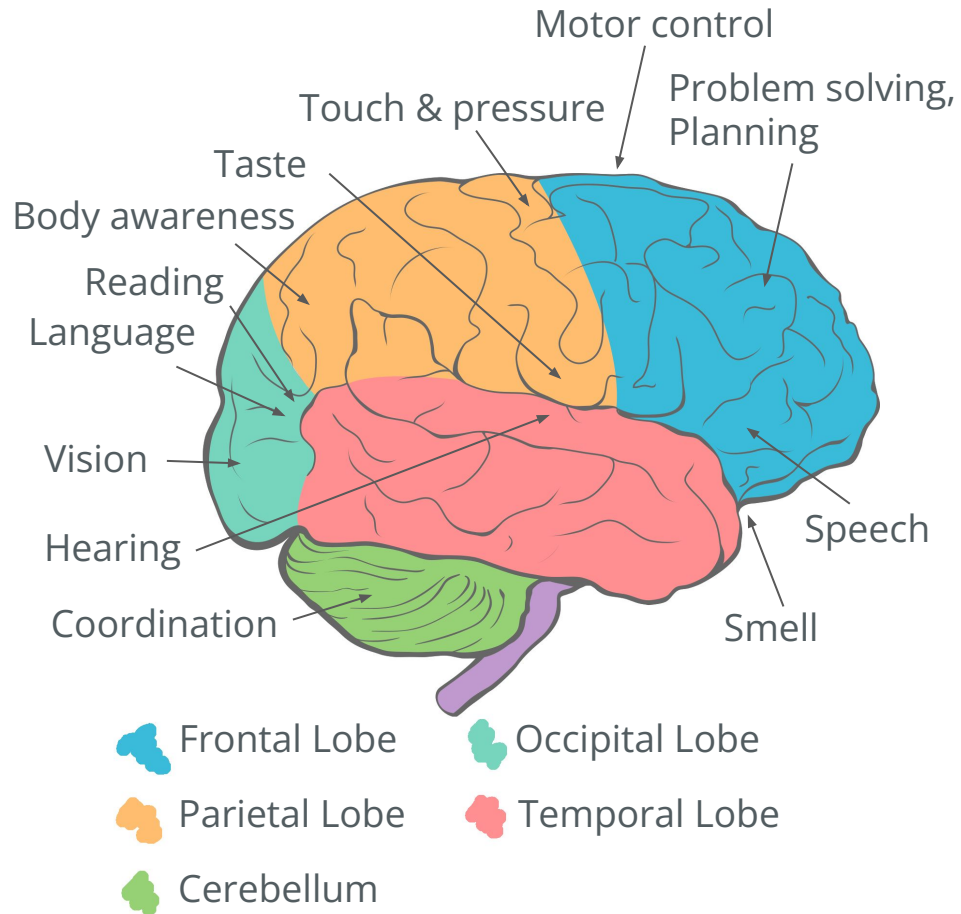
- Bot/dots - [video](#) - lol - [video](#)
- **Input/output:** voice, video, code, control actions (tokens)



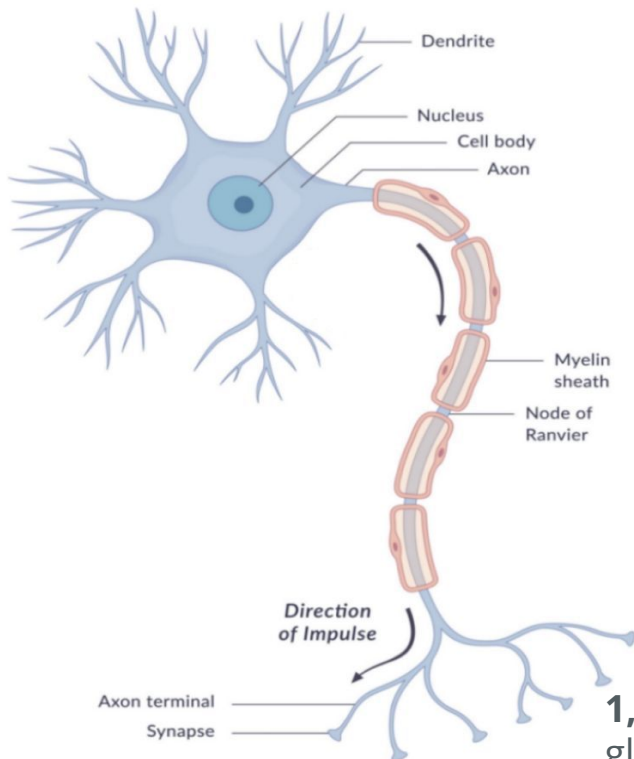
Architecture of the brain

Right side/left side (hemispheres)

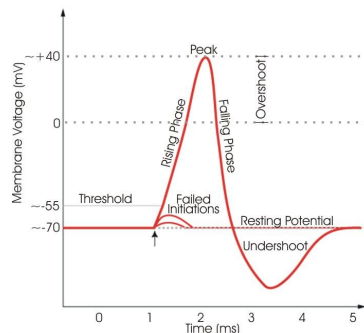
- Frontal lobe
 - executive functions, memory and planning
- Parietal lobe
 - sensation and spatial awareness
- Temporal lobe (banana shape)
 - hearing and language
- Occipital lobe at back
 - Vision from front along optic nerves



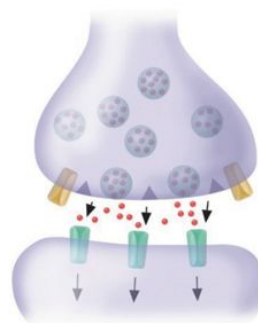
1,000's of inputs (other neurons, sensory neurons e.g. taste buds from a salt or sugar molecule...)



- Neurons send out branches called **dendrites**, and a large output called an **axon**
- The axon is coated in myelin that helps it conduct electrical impulses
- The places where the nerve cells make their connections with each other is called a **synapse**

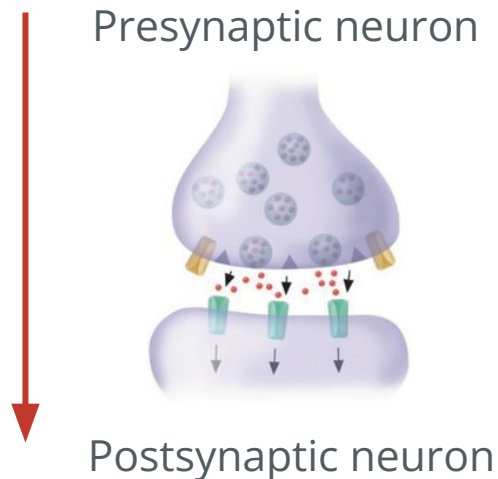


Synapse:

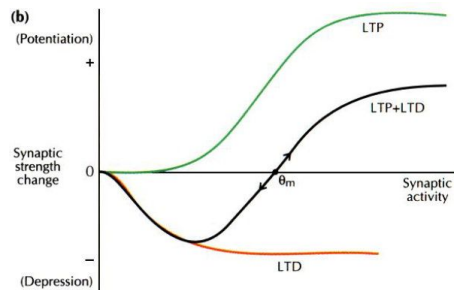


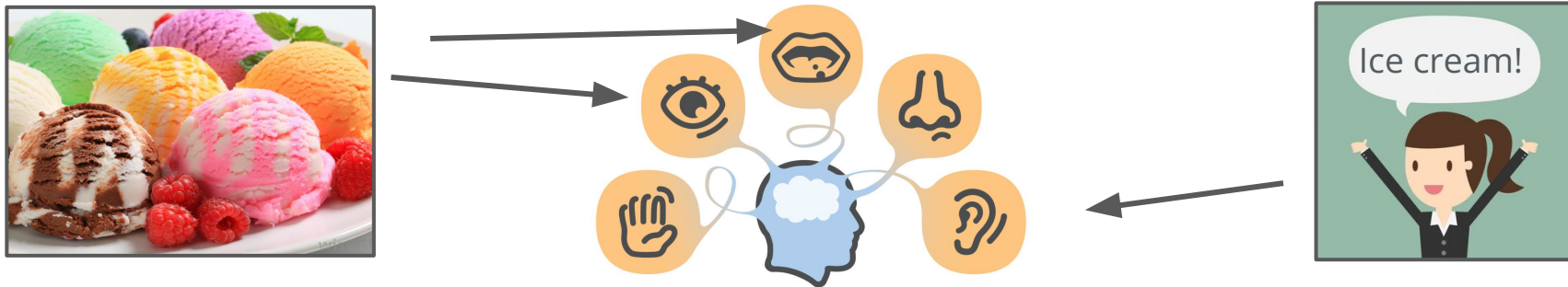
- "Synapse" (from the greek meaning "to clasp together")
- **Signals** get **summed up**, and travel to the hillock (Axon neck)
 - If large enough, triggers an action potential travels down axon

1,000's of output targets (e.g. other neurons, muscle cells, gland cells, blood vessels to release hormones...)



- What's very cool is that with frequent repeated stimulation, the same level of presynaptic stimulation converts into **greater** postsynaptic potential
 - In other words, as a neuron gets a lot of practice sending signals to a specific target neuron, it gets better at sending those signals (the synapse strength increases)
 - Increased strength that lasts for a long time (from minutes to many months) is called **Long Term Potentiation** (weakening is **Long Term Depression**)
 - As synapses are strengthened and retain strength, we're able to more easily recall previous experiences



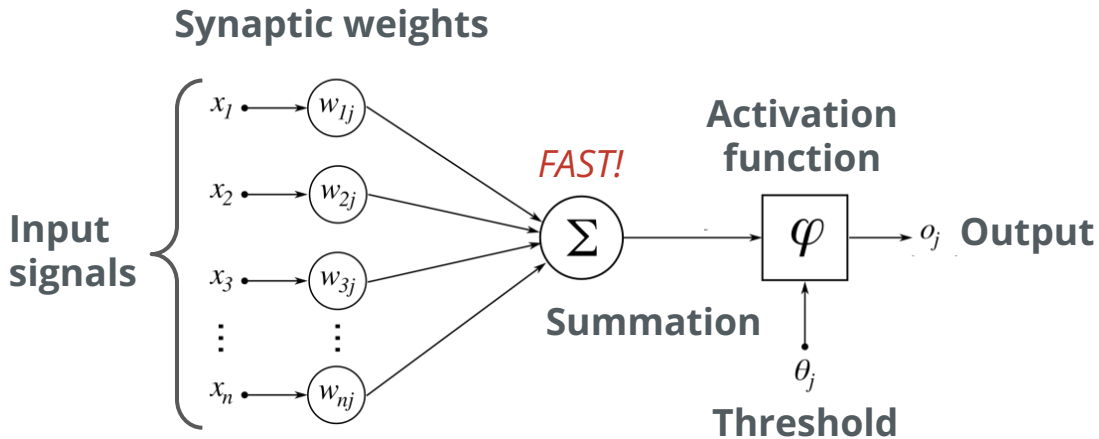
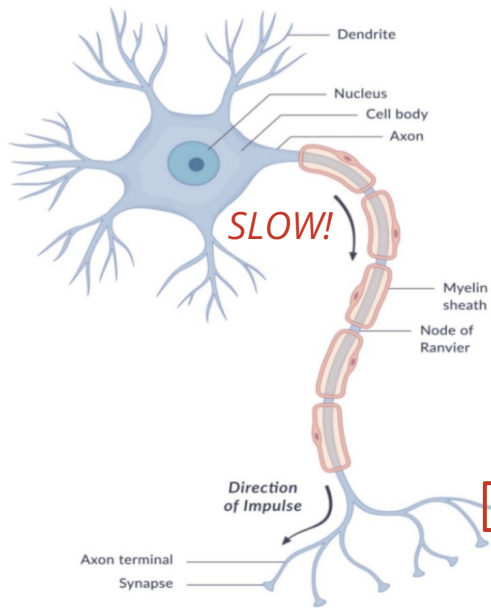


- Hebbian theory
 - If two neurons fire at the same time, the connections between them are strengthened, and thus are more likely to fire again together in the future
 - If two neurons fire in an uncoordinated manner, their connections are weakened, and they're more likely to act independently in the future
- Updated Hebbian hypothesis based on recent findings
 - If the presynaptic neuron fires within a window of 20ms before the postsynaptic neuron, the synapse will be strengthened
 - However if within a window of 20ms after, the synapse will be weakened

A brief history artificial neurons

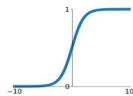
Pitts and McCulloch, 1943

The Artificial Neuron



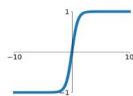
Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



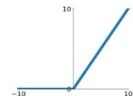
tanh

$$\tanh(x)$$



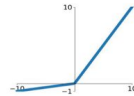
ReLU

$$\max(0, x)$$



Leaky ReLU

$$\max(0.1x, x)$$

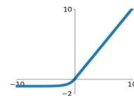


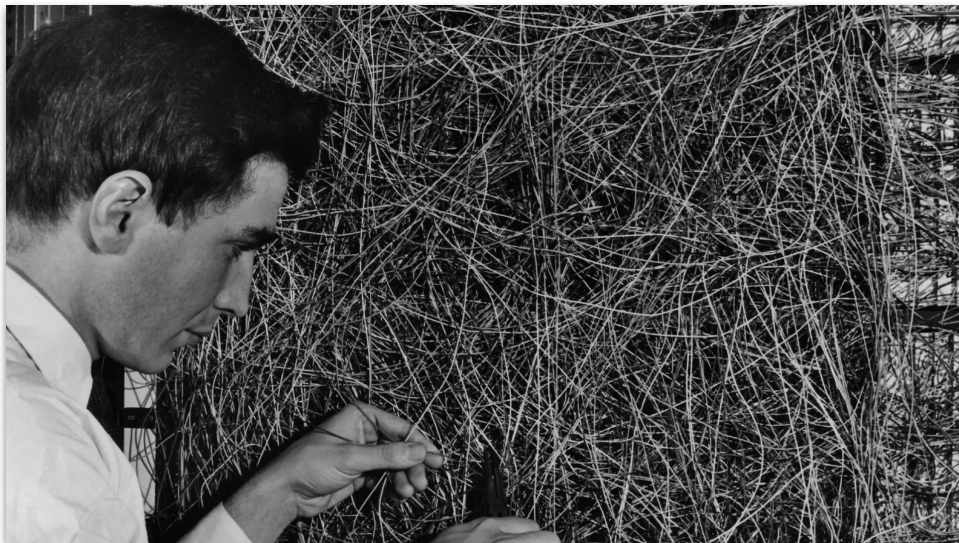
Maxout

$$\max(w_1^T x + b_1, w_2^T x + b_2)$$

ELU

$$\begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases}$$





Frank Rosenblatt, 1957

The perceptron machine was difficult to tune

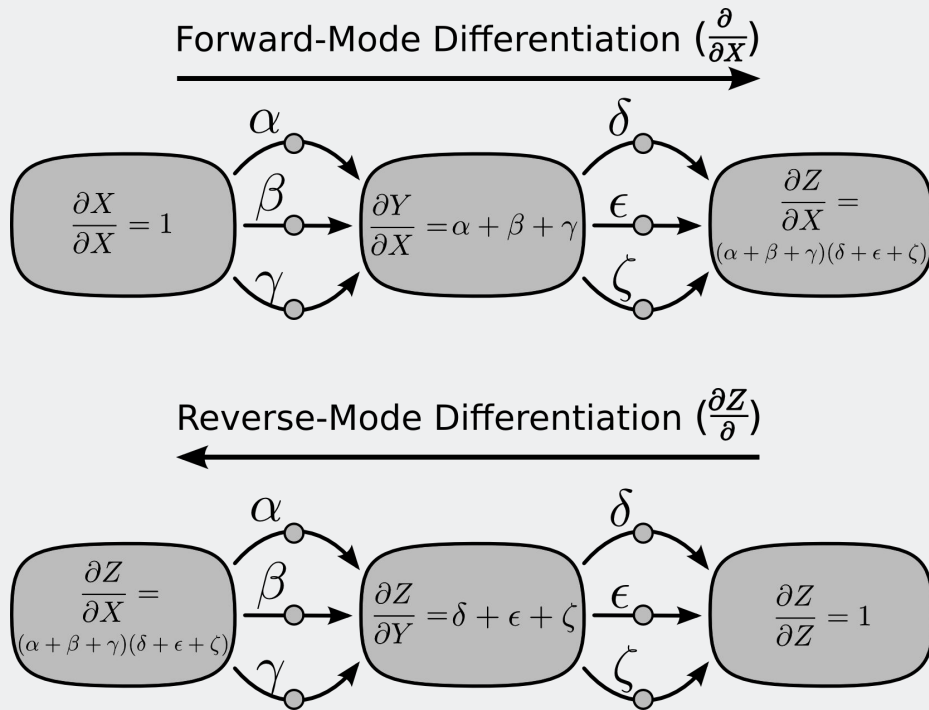


Backpropagation, 1970

The backpropagation algorithm (also called the reverse mode of automatic differentiation) was independently discovered by different researchers

Seppo Linnainmaa, 1970
Werbos, 1974 (with neural networks)

Reverse-mode differentiation



Rumelhart and Hinton, 1986

Learning representations by back-propagating errors

Nature, 1986. David E. Rumelhart, Geoffrey E. Hinton & Ronald J. Williams

- Backpropagation
- Multiple hidden layers
- Recurrent networks

The second AI winter 1987–1993





History of CNNs

Convolutional neural networks were first introduced by Kunihiro Fukushima in 1980

- **1989** - Yann Lecun et al., trained a CNN with “Backpropagation Applied to Handwritten Zip Code Recognition”
- **1998** - Yann Lecun et al., released LeNet5 “Gradient-based learning applied to document recognition”

Example: LeNet5

```
class LeNet(nn.Module):  
    def __init__(self):  
        super(LeNet, self).__init__()  
        self.conv1 = nn.Conv2d(1, 6, 5, padding=2)  
        self.conv2 = nn.Conv2d(6, 16, 5)  
        self.fc1 = nn.Linear(16*5*5, 120)  
        self.fc2 = nn.Linear(120, 84)  
        self.fc3 = nn.Linear(84, 10)  
  
    def forward(self, x):  
        x = F.max_pool2d(F.relu(self.conv1(x)), (2, 2))  
        x = F.max_pool2d(F.relu(self.conv2(x)), (2, 2))  
        x = flatten(x)  
        x = F.relu(self.fc1(x))  
        x = F.relu(self.fc2(x))  
        x = self.fc3(x)  
        return x
```



The 2010 AI spring



History of NNs on GPUs

- **2004** - first GPU implementation of a neural network
- **2006** - first GPU implementation of a CNN (just 4 times faster)
- **2012** - AlexNet - won state-of-the-art by significant margin with 60 million parameters
- **2015** - ImageNet state-of-the-art by a residual network with over 100 layers

Cloud Computing - GANs & GPT-4

Fake faces generated by StyleGAN2



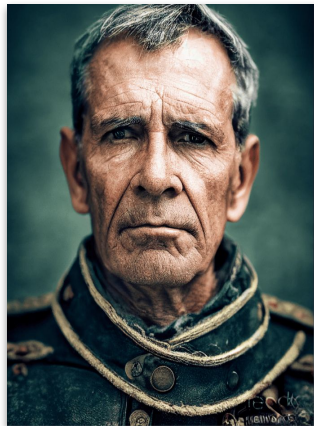
Unsupervised Generative Models

Progressively larger models are being trained on GPU cloud services

- **2014** - Ian Goodfellow releases "Generative Adversarial Nets"
- **2017** - GoogleBrain releases "Attention is all you need"
- **2019** - OpenAI GPT-2 "Language Models are Unsupervised Multitask Learners"
- **2020** OpenAI GPT-3 "Language Models are Few-Shot Learners"

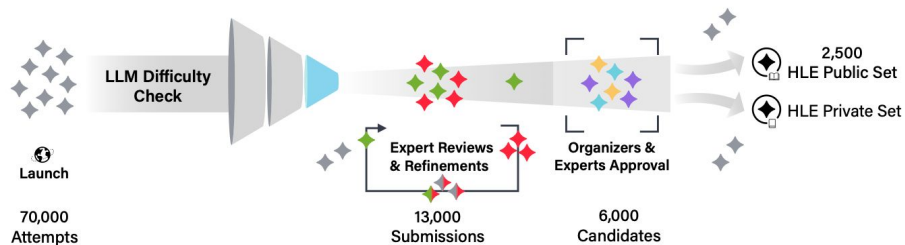
2023

GPT-4 multimodality, stable diffusion, plugins...

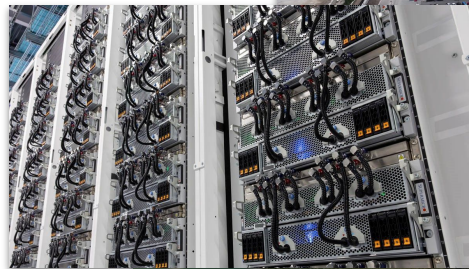


"Cyberpunk style image of a motorcycle reflection in rain, ukiyo-e, unreal engine, trending on artstation"

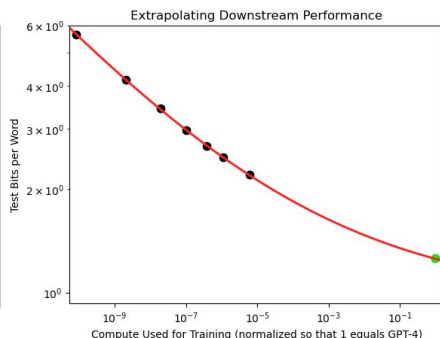
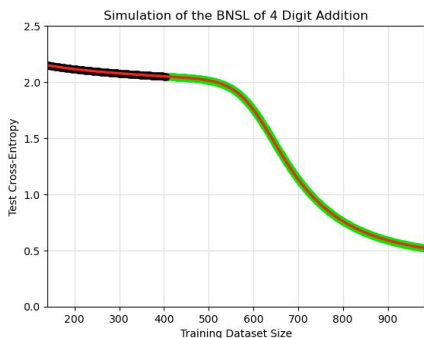
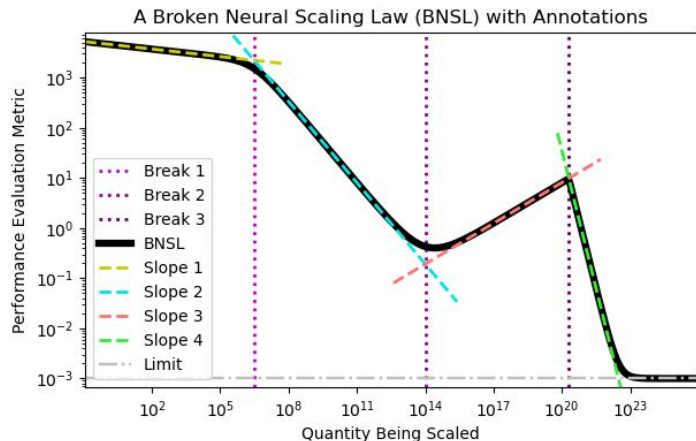
2026+ ...to context engineering, agents, tool use, pay-to-win



...building in progress...



https://scale.com/leaderboard/humanitys_last_exam



Neural scaling laws

Can make excellent predictions of future generalisation performance with more £.

- **2017** - Hestness et al., “Deep learning scaling is predictable, empirically” (when first started teaching DL at Durham!)
- **2023** - “Broken Neural Scaling Laws” captures double descent etc
- **2024+** “Scaling LLM Test-Time Compute Optimally can be More Effective than Scaling Model Parameters”

The reasoning debate where are you?



Several of these people's views have changed over the past 5 years...

Solving tasks $\neq$ reasoning

Predicting the future is intelligence



LLMs don't understand the world

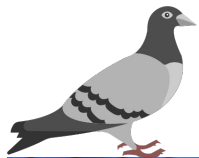
We're not so different

“Stochastic parrots... do not understand” + long-term value optimisation Superintelligence

The safety & alignment debate where are you?

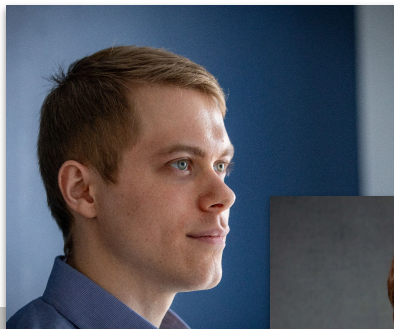


Address immediate
real harms



Stop exaggerating

Policy and strategy,
an OFF switch &...



Safety and
regulation



We need better
AI reward design

We're screwed!



Existential risk



Machine Learning

Reinforcement Learning

bad gradients
dynamic not IID
optimising future reward
surprising

Multi Agent Learning

solving tasks together
decentralized societies
agent economies

Deep Learning

multiple layers
good gradients
mostly IID

Unsupervised Learning

generalise to different
data (hard*)



Generative Models

learning the task/data
distribution

Shallow Learning

often hand-engineered

Supervised Learning

generalise to similar data
(solved*)



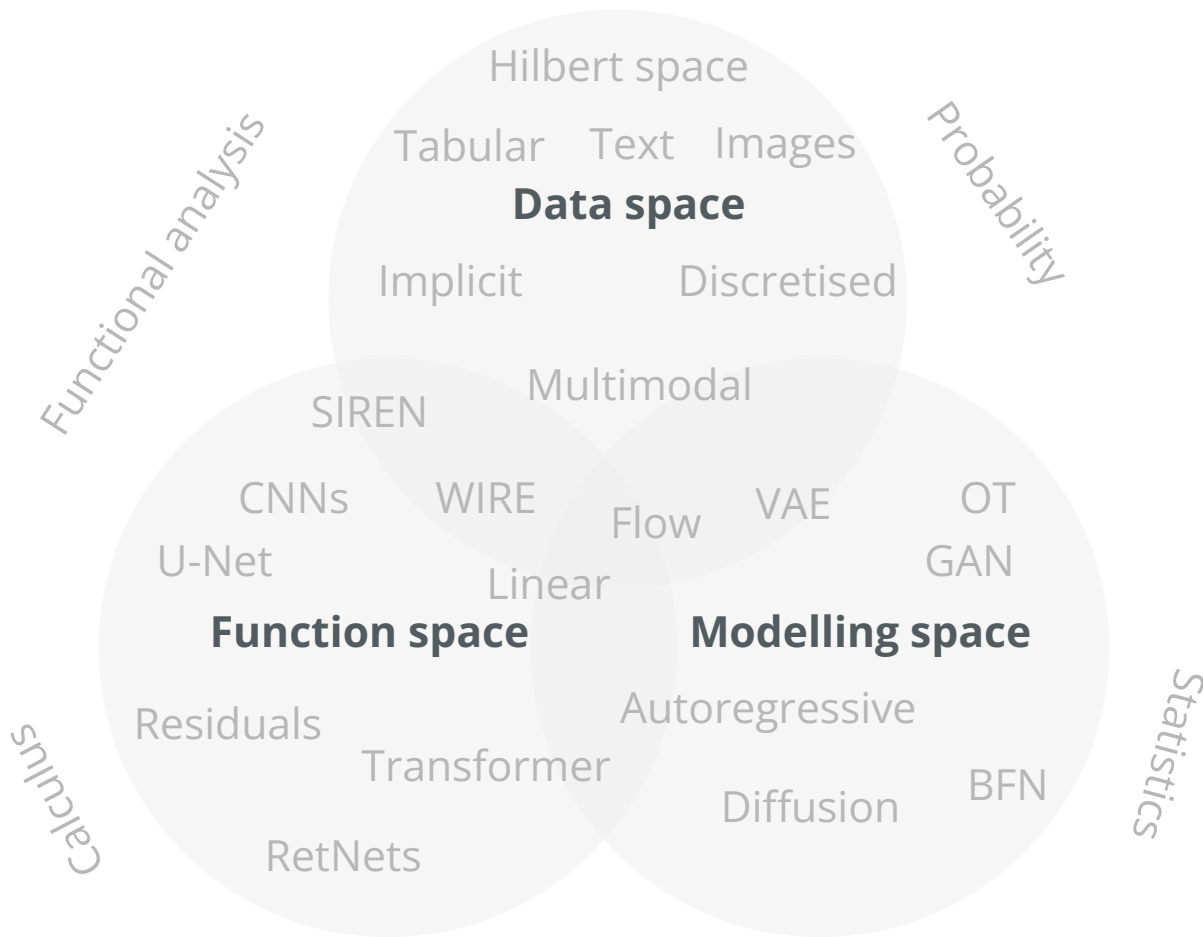
Discriminative Models

classifying data
detecting regions
paired-translation

...

Key concepts

deep learning contributions from three spaces





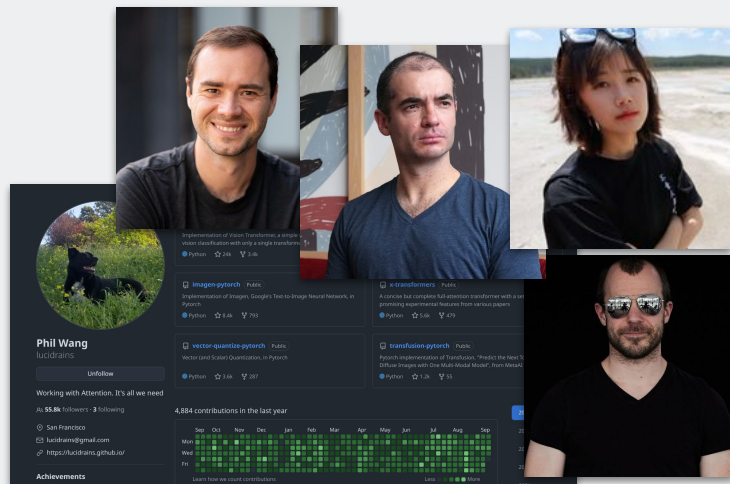
Follow key resource hubs & people

1. <https://www.latent.space/>
2. Reddit, discord, etc
r/MachineLearning/
MLOps discord
3. Top conferences



Key educators, devs & companies

Andrej Karpathy, Lilien Weng, Yannic Kilcher, lucidrains, AK, Hendrycks, ...



Summary

In summary, deep learning:

- has overlap with many areas
- achieves state-of-the-art in ill-defined tasks and scales with...
 - very high-dimensional datasets
 - huge datasets
 - parallelizable hardware
- fields of maths: probability, calculus, statistics, functional analysis...
- is rapidly growing and evolving